Module 1 Homework 4 Challenge

Below is a scored review of your assessment. All questions are shown.

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1

Who gets more work done, $2x$ people who each work at a rate of $\frac{1}{2}\frac{\text{job}}{\text{hour}}$, or x people who each work at a rate of $1\frac{\text{job}}{\text{hour}}$?

Your Answer	They get the same amount of work done in the same amount of time
Correct Answer	They get the same amount of work done in the same amount of time

Explanation

The $2x$ people are twice as numerous as the x people, but their work rate is half that of the x people. Then, their work rate would be $2 \times \frac{1}{2}$ the rate of the x people, which simplifies to $1\ \frac{\text{job}}{\text{hour}}$. Therefore, they work at the same rate, and choice three is the correct answer.

2

140 pounds of fresh cherries can be dried to get 21 pounds of dried cherries. How many pounds of dried cherries can be gotten from 160 pounds of fresh cherries?

Your Answer	24
Correct Answer	24

Explanation

We know that 140 pounds of fresh cherries turns into 21 pounds of dried cherries, and we want to know how many pounds of dried cherries we can get from $160\ \mathrm{lb}$ fresh cherries. The ratio of fresh cherry to dried cherry is $\frac{140}{21} = \frac{20}{3}$. Since $160 = 20 \times 8$, the amount of dried cherries we can get will be 20 times 3, or 24 pounds.

3 What would be the rate of water flow of an intake pipe that can fill a swimming pool in 3 hours?

Your Answer	$\frac{1}{3}\ \frac{\text{pool}}{\text{hour}}$
Correct Answer	$\frac{1}{3}\ \frac{\text{pool}}{\text{hour}}$

Explanation

We aren't given any information regarding the number of gallons that the pipe intakes for a given unit of time. Therefore, the answer must involve only the pool and the 1 hour. The rate would give the relationship between the pool and the amount of time to fill it. A larger rate indicates a faster amount of work done per unit time, so the third choice wouldn't work. If the third choice were larger in size, that would mean it takes more time to fill the pool, which would actually mean a smaller rate. The second choice agrees with this, however, since if this fraction is larger, that means more pool gets filled with each unit time. The second choice is the correct answer.

4 Which of the following is FALSE about the relationship between amount of work done, number of people doing the
work and the amount of time
spent doing the work?

Your Answer	The less work that is done, the more time it takes
Correct Answer	The less work that is done, the more time it takes

Explanation

We just have to remember the relationships between each pair of quantities: work and time, work and people, and time and people. Work and people increase together, and work and time also increase together, but time and people are inversely related. If you have more people, this reduces the time it takes to accomplish something, since all the people work together. All the choices match these relationships except for the fourth one, which says that the less work that is done, the more time it takes. Since the question says that all are true except for one, the fourth choice is the choice we are looking for.

5 A company wants to make 250 protractors and 60 calculators every hour. A given worker can either make 10 protractors or 4 calculators per hour. How many workers will the company need to make the exact amount of products every hour?

Your Answer	40
Correct Answer	40

Explanation

The company wants 250 protractors an hour, where each worker can make 10 each hour, so we need $\frac{250}{10} = 25$ workers making protractors. Similarly, in order to make 60 calculators per hour, they will need $\frac{60}{4} = 15$ workers, as each person can make four calculators in a given hour. Add the two numbers, and we get $25 + 15 = 40$ workers.

6

A large water tank has two inlet pipes (a large and a small one) and one outlet pipe. It takes 2 hours to fill the tank with the large inlet pipe. On the other hand, it takes 5 hours to fill the tank with the small inlet pipe. The outlet pipe allows the full tank to be emptied in 7 hours. Assuming that the tank is initially empty, what fraction will be filled in 1 hour if all three pipes are in operation? Your answer should be a fraction in simplest form, without spaces, e.g. 1/2.

Your Answer	39/70
Correct Answer	39/70

Explanation

Let V be the volume of the tank. The large inlet pipe has a flow rate of $\frac{V}{2}$, the small inlet pipe has a flow rate of $\frac{V}{5}$, and the outlet pipe has a flow rate of $-\frac{V}{7}$. Thus, the total flow rate with all three open is $\frac{V}{2} + \frac{V}{5} - \frac{V}{7} = \frac{39V}{70}$. We can clearly see that the fraction of V that will be filled is $\frac{39}{70}$.

7

Alice and Bob start from the same position on a track and both start jogging clockwise along it. Alice runs at 4 miles per hour, and Bob runs at 3 miles per hour. How much extra distance will Alice have covered when she passes Bob?

Your Answer	One track lap
Correct Answer	One track lap

Explanation

Alice covers more distance than Bob does, because she is faster, but since when she passes Bob they are at the same point on the track, that means that the extra distance she has gone compared to him is an integer multiple of track lengths. Since this is the first time that she passes Bob, this must equal 1 track length, so the third choice is the correct answer.

8 You just finished a triathlon in a spectacular 2 hours, and you want to know how fast you swam. The race consists of 1 mile of swimming, 12 miles of bicycling, and 3 miles of running. You know that you biked for 1 hour and you ran at a speed of 5 miles per hour. How many miles per hour did you swim? Your answer should be a number only, rounded to the nearest tenth, without units.

✓	**Your Answer**	2.5
	Correct Answer	2.5

Explanation

To get your swimming speed, we need the distance you swam, and also the time it took to swim that distance. We are given the distance, but not the time. Since biking took 1 hour, you know that the running and swimming combined also took you 1 hour. You ran at a speed of 5 miles per hour for 3 miles, so that took $\frac{3}{5} = 0.6$ hours, or 36 minutes. That means you swam for 24 minutes, or 0.4 hours, so computing distance over time, we get a swimming rate of $\frac{1}{0.4} = 2.5$ miles per hour.

9 The harmonic mean, a type of average of two speeds, is different from simply the arithmetic average of two rates, because

Your Answer	[no answer given]
Correct Answer	If you go equal distance at speed number one and the same distance at speed number two, you don't spend equal amounts of time at each speed, since the faster speed will get you there in less time

Explanation

The harmonic mean was derived in the Day 15 lesson. It really means the average speed, defined as

$$\frac{\text{total distance}}{\text{total time}}$$

The key here is that you travel the same distance at one speed, let's call it v_1, as you do at a different speed, v_2. Let's call this distance d. The time you spend at speed 1 is

$$\frac{d}{v_1}$$

and the time you spend at speed 2 is

$$\frac{d}{v_2}.$$

From this, we see that the two are not equal, so simply taking the sum of the speeds and dividing by two wouldn't be correct. The harmonic mean should be closer to the speed that you spend more time at. The correct answer is the first choice.

The second choice is incorrect for two reasons; a mean and an average both mean the same thing, the arithmetic mean, which is referred to in choice four. The third choice is wrong because the median is the value that lies in the middle position of all the numbers if they are ordered from smallest to largest. The fifth choice says that the harmonic mean is the simple arithmetic average of the speeds, which isn't true, as you can see in the math note below.

Math note. In the videos, we showed how to find a formula for the harmonic mean. It was the reciprocal of the sum of the reciprocals of the speeds. To get this, start with finding average speed:

$$
\begin{aligned}
\frac{\text{total distance}}{\text{total time}} &= \frac{d+d}{\frac{d}{v_1}+\frac{d}{v_2}} \\
&= \frac{2d}{\frac{d}{v_1}+\frac{d}{v_2}} \\
&= \frac{2}{\frac{1}{v_1}+\frac{1}{v_2}} \\
&= \frac{2}{\frac{v_1+v_2}{v_1\times v_2}} \\
&= \frac{2v_1v_2}{v_1+v_2}
\end{aligned}
$$

10 If a ants can dig b tunnels in c hours, how many tunnels can b ants dig in a hours?

✕	**Your Answer**	[no answer given]
	Correct Answer	$\frac{b^2}{c}$

Explanation

Since a ants can dig b tunnels in c hours, a ants can dig $\frac{b}{c}$ holes each hour. Then 1 ant can dig $\frac{b}{ca}$ tunnels per hour. Then, if there are b ants and a hours, the total number of tunnels is the number of tunnels per hour times the number of ants times the number of hours, which is $\frac{b}{ca} \times ba$, which equals $\frac{b^2}{c}$.

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 A train travels at a constant speed of 120 meters per second. The front of the train has just reached the entrance of a tunnel. Two seconds later, all of the train is inside the tunnel, and 3 seconds after that, the back of the train exits the tunnel. What is the length of the train divided by the length of the tunnel? Your answer should be a fraction in simplest form, without spaces, e.g. 1/2

Your Answer	[no answer given]
Correct Answer	2/3

Explanation

When the front of the train has been in the tunnel for 2 seconds, the back of the train enters the tunnel. This means it takes 2 seconds from the time the front of the train passes a certain point, to when the back of the train does so. Therefore the length of the train is $120 \times 2 = 240$ meters.

From the point when the back of the train enters the tunnel to the point when the back of the train exits the tunnel, the time that has elapsed is 3 seconds. This means the tunnel takes 3 seconds to completely go through, which means the tunnel is $120 \times 3 = 360$ meters long.

Therefore, the ratio of the train's length to the tunnel's length is

$$\frac{240}{360} = \frac{2}{3}.$$

12 A cyclist travels at a constant speed for 6 hours over a distance d. How much longer, in hours, will it take for him to travel if he goes 50% longer distance? Your answer should be the number only, without units.

Your Answer	[no answer given]
Correct Answer	3

Explanation

Let that the distance that the cyclist travels in 6 hours be d. Then his new distance will be $1.5 \times d$. The cyclist goes at a constant speed which is equal to

$$\frac{\text{total distance}}{\text{total time}} = \frac{d}{6}.$$

Now, let's find the trip time for the new distance.

$$\text{new trip time} = \frac{\text{newdistance}}{\text{speed}} = \frac{1.5 \times S}{\frac{S}{6}} = 1.5 \times 6 = 9.$$

The difference in trip times is then $9 - 6 = 3$ hours.

13 The long hand of a clock takes 60 minutes for one rotation, while the small hand takes 12 hours for one rotation. If it is $2:00$ p.m. now, at what time will the two hands overlap for the 5th time?

✕	**Your Answer**	[no answer given]
	Correct Answer	6:33 p.m.

Explanation

Each hour, the small and long hand will overlap once, so they will overlap at $2:xx$ p.m., $3:xx$ p.m., etc. The 5th overlap then will be at $6:xx$ p.m. At $6:00$ p.m., the small hand will be pointing down to 6 o'clock, while the long hand will be pointing up to 12 o'clock. We can now convert this to a linear equation. Let the small hand move at a rate of x, then the long hand will move at a rate of $\frac{x}{12}$, as it takes 12 hours to make one rotation. The long hand also starts pointing down, which also can be interpreted as 30 minutes, so we get the following equation $x = \frac{x}{12} + 30$ as we want to find out when they are pointing to the same value. Solving the equation gives us $x = 32.73$ so the closest minute value to that is 33, so we get $6:33$ p.m.

14 A car travels at a speed of 15 kilometers per hour (kph) for 4 hours, then travels for a distance of 30 kilometers at a speed of 20 kph, then 144 kilometers at 36 kph, and 12 kph for the last 30 minutes. What was the kilometers per hour of the car's average speed?

Your Answer [no answer given]

Correct Answer 24

Explanation

Average speed is just $\frac{\text{total distance}}{\text{total time}}$, so we can just add up our total distance and total time. We can use the fact that $\text{time} = \frac{\text{distance}}{\text{speed}}$.

$$\begin{aligned}
\text{total distance} &= 15\frac{\text{km}}{\text{hr}} \times 4\text{hours} + 30\text{km} + 144\text{km} + 12 \text{ kph} \times \frac{1}{2} \text{ hour} \\
&= 60 + 30 + 144 + 6 \text{ km} \\
&= 240 \text{ km} \\
\text{total time} &= 4\text{hours} + \frac{30 \text{ km}}{20 \text{ kph}} + \frac{144 \text{ km}}{36 \text{ hours}} + \frac{1}{2} \text{ hours} \\
&= 4 + 1\frac{1}{2} + 4 + \frac{1}{2} \\
&= 10 \text{ hours} \\
\text{average speed} &= \frac{240 \text{ km}}{10 \text{ hours}} \\
&= 24\frac{\text{km}}{\text{hr}}.
\end{aligned}$$

15 A car is driving at 40 mph. It passes a train station. Fifteen minutes later, a train passes through the station at 60 mph. How far from the train station will the car be when it is passed by the train, if both vehicles stay at a constant speed the whole time? Your answer should be the number only, without units.

Your Answer	[no answer given]
Correct Answer	30

Explanation

Fifteen minutes after the car passes the train station, the car is $\frac{40 \times 15}{60} = 10$ miles away from the station. The train is going at 60 miles per hour. To get the time it takes the train to catch up to the car, we can say that the train is going at 20 miles per hour relative to the car, so if the car is 10 miles ahead, it takes 30 minutes to catch up to the car. In that time, the distance from the station is $10 \text{ miles} + \frac{40 \times 30}{60} = 30 \text{ miles}$ away from the station.

16 Elizabeth leaves home at $7:30$ p.m., walking at 3 miles per hour. Once she has gone 4 miles, Robert leaves the same house and bikes on her path at 19 miles per hour. How many minutes later after $7:30$ p.m. do Elizabeth and Robert meet? Your answer should be the number only, without units.

✕	**Your Answer**	[no answer given]
	Correct Answer	95

Explanation

When Robert leaves the house, Elizabeth has gone 4 miles, and since her speed is 3 miles per hour, she can go 1 mile in $\frac{1}{3}$ of an hour. This means she had been gone for $1+\frac{1}{3}$ hour, which is 1 hour and 20 minutes. When Robert leaves, then, the time is $7:30$ plus 1 hour and 20 minutes, which is $8:50$ p.m. Every hour, Robert catches up by 16 miles, because his speed is faster by $16-3$ miles per hour. Therefore, it will take him $\frac{1}{4}$ hour, or 15 minutes, after $8:50$ p.m., to catch up. Then the time it takes for him to catch up is 1 hour and 20 minutes plus 15 minutes, which is 95 minutes.

17 Suppose you take a trip with your car and drive uphill for 40 miles, which uses up $\frac{1}{6}$ of your gas tank, and then you drive your car downhill for 60 miles, which uses up another $\frac{1}{6}$ of your tank. You arrive at Numberland and have a day of math fun. Before you leave to go home, you fill your tank with 2 gallons of gas, and then you make the trip back using the same route. Assuming that your tank has a capacity of 12 gallons and you left home with a full tank, how many gallons of gas will you have when you get back home? Your answer should be a decimal, rounded to the nearest tenth, without units.

Your Answer	[no answer given]
Correct Answer	5.7

Explanation

Let's start by calculating the rate of gas used going uphill and downhill. You went uphill for 40 miles, which used $\frac{1}{6}$ of your tank, which is 2 gallons, so you traveled 40 miles per 2 gallons, or in other words, 20 miles per gallon. Going downhill 60 miles also took you $\frac{1}{6}$ of your tank, which is 2 gallons, so you went 60 miles per 2 gallons, or 30 miles per gallon. For the trip back, you went 60 miles uphill, which used $\frac{60}{20} = 3$ gallons, and 40 miles downhill, which used $\frac{40}{30} = \frac{4}{3}$ gallons. Add them together, and the total amount of gas used to go back is $4\frac{1}{3}$ gallons. Counting in the full tank of 12 gallons we had at the beginning, the two gallons we added in the middle, and the wo trips, we get

$$12 - 4 + 2 - 4\frac{1}{3} = 5\frac{2}{3},$$

which is closest to 5.7.

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 A pirate tries to explode an enemy ship by lighting a fuse in the ammunition hold. Every minute, $\frac{1}{4}^{\text{th}}$ of the fuse burns away. He spends 16 seconds running through the ship, which is 10 meters long, and then escapes into a jolly-boat which he rows to shore 20 meters away. His rowing speed is one-fifth of his running speed, and if he brings n treasure chests onto the jolly-boat, his rowing speed reduces by $\frac{n}{10+n}$ of his original rowing speed. How many treasure chests can he carry to shore in one trip before the fuse detonates the explosives?

✕	**Your Answer**	[no answer given]
	Correct Answer	4

Explanation

There are a lot of different rates in this problem, but we can tackle them one at a time. Since the rate of the fuse burning seems to not be related to the other rates, we can think about that one first. If one-fourth of the fuse burns every minute, it will take $4 \times 1 = 4$ minutes for the fuse to burn. The other times are given in seconds, so let's convert the 4 minutes into 240 seconds.

Now let's think about how much time it takes him to row to shore. We know the distance to shore is twice the distance he ran through the ship, and his rowing speed is also one-fifth his running speed, so the time it will take to row to shore is 10 times the time he runs through the ship, which is 10×16 seconds. This means that without carrying any treasure chests, it will take him $10 \times 16 = 160$ seconds to row to shore.

The time it takes to go through the ship and to row is $160 + 16 = 176$ seconds. Since he has 240 seconds in total, the extra time he has to spare is $240 - 176 = 64$ seconds. This is the time that can be used towards carrying extra chests.

If he brings n chests, his speed reduces by $\frac{n}{10+n}$ of the original speed, and accordingly his rowing time increasese by $\frac{n}{10+n}$ of his original rowing time.

Let's see why this is true! If he brings n chests, and letting $v_{\text{no chests}}$ be his speed without any chests, his new speed, $v_{\text{with n chests}}$, becomes

$$\begin{aligned} v_{\text{with n chests}} &= v_{\text{no chests}} - \left(\frac{n}{10+n}\right) \times v_{\text{nochests}} \\ &= \left(\frac{10+n}{10+n}\right) \times v_{\text{no chests}} - \left(\frac{n}{10+n}\right) \times v_{\text{no chests}} \\ &= \left(\frac{10}{10+n}\right) \times v_{\text{no chests}} \end{aligned}$$

With this new speed, we can use the rate equation to find his new time, $t_{\text{with n chests}}$. The time it takes with no chests is:

$$\begin{aligned} d &= v_{\text{no chests}} \times t_{\text{no chests}} \\ \implies t_{\text{no chests}} &= \frac{d}{v_{\text{no chests}}} \end{aligned}$$

and the time it takes with n chests is:

$$t_{\text{with n chests}} = \frac{d}{v_{\text{with n chests}}}$$

Using our expression for $v_{\text{with n chests}}$ from earlier, we get

$$t_{\text{with n chests}} = \frac{d}{\frac{10}{10+n} \times v_{\text{no chests}}}$$

And replacing $\frac{d}{v_{\text{no chests}}}$ with $t_{\text{no chests}}$, we can get

$$t_{\text{with n chests}} = \frac{1}{\frac{10}{10+n}} \times t_{\text{nochests}}$$

$$t_{\text{with n chests}} = \left(\frac{10+n}{10}\right) \times t_{\text{nochests}}$$

which means the time increases by $\frac{n}{10+n}$ of the original rowing time.

His original rowing time was 160 seconds, and the extra rowing time is 64 seconds, which is $\frac{64}{160} = \frac{16\times4}{16\times10} = \frac{4}{10}$ of the original rowing time. Thus $n = 4$ and the pirate can carry $\boxed{4}$ treasure chests without sinking before he reaches the shore.

19

Train stations A and B are connected by a straight track that is 72 miles long. A slow freight train traveling at 18 miles per hour leaves A and heads towards B. At the same time, a bird who flies at 30 miles per hour leaves B and heads towards A. When the bird meets the train, it turns around "instantaneously" (that is, imagine it gets reflected) and heads back towards B. When the bird reaches the station, it turns around and heads back towards A until it meets the train. Then the bird then turns around again and heads back towards the station. This process continues until the train reaches B. What is the total distance flown by the bird?

Your Answer	[no answer given]
Correct Answer	120

Explanation

We can essentially ignore the bird turning around and imagine that it instead flies continuously straight. It takes the train $\frac{72}{18} = 4$ hours to reach the opposite station, and during this time, the bird will fly $30 \cdot 4 = 120$ miles.

20 Each sheet of paper costs $\$0.0025$ and each cardboard cover costs $\$0.20$. If a notebook is made using 168 sheets of paper and two cardboard covers, how many **dollars** does it cost to manufacture 200 notebooks?
Your answer should be the number only, without units.

Your Answer	[no answer given]
Correct Answer	164

Explanation

Solution 1. One step at a time. For one notebook, we need 168 sheets of paper and 2 covers, which would cost

$$168 \times 0.25 + 2 \times 20 \ \text{ cents}$$

Since $0.25 = \frac{1}{4}$, the expression becomes

$$\begin{aligned} 168 \times \frac{1}{4} + 2 \times 20 &= 42 + 2 \times 20 \\ &= 82 \ \text{ cents} \end{aligned}$$

We want the cost of 200 notebooks, so we want

$$200 \times 82 = 100 \times 2 \times 82 = 16400 \text{ cents}$$

This is equal to $\$164$.

Solution 2. String all the calculations together. From the video lessons, we saw it can be cleaner to delay calculating until the very last step, after you have combined all of the operations that you need. In this case, we should multiply the cost of one notebook,

$$168 \times 0.25 + 2 \times 20 \ \text{ cents},$$

by the number of notebooks, 200 :

$$200 \times (168 \times 0.25 + 2 \times 20) \ \text{ cents},$$

and then divide by 100 to convert from cents to dollars:

$$\frac{1}{100} \times 200 \times (168 \times 0.25 + 2 \times 20) \ \text{ dollars}$$

The 100s cancel out, leaving you with

$$2\,(168 \times 0.25 + 2 \times 20) \ \text{ dollars}$$

which is a tiny bit simpler than the version in the first solution.

Summary

Software by
brillium
Version 11.2

Privacy Policy. Assessment content is copyright 2026, Expii, Inc..